CoCo Desktop — Dev Day

Hands-On Technical Training for Kroger Health & Wellness

Developers, Data Engineers & Data Scientists • 3-Hour Session • October 14, 2026

CREATED WITH **SNOWFLAKE COWORK**

Training Overview

Session Title: CoCo Desktop for Developers, Data Engineers & Data Scientists

Duration: 3 Hours (Hands-On Lab)

Date: October 14, 2026

Audience: H&W Developers, Data Engineers & Data Scientists

Format: Instructor-led overview + guided lab, then self-paced choose-your-own-adventure labs

Goal: Enable technical users to leverage CoCo Desktop for data exploration, pipeline building, query optimization, app development, and AI/ML workflows within Snowflake.

Learning Objectives

By the end of this session, participants will be able to:

- Navigate CoCo Desktop and apply best practices for AI-assisted development
- Explore data, generate queries, and build pipelines using natural-language prompts
- Create Semantic Views, Cortex Agents, and CoCo Skills
- Optimize slow queries and diagnose performance issues with AI assistance
- Build Streamlit apps, Notebooks, and data-quality checks directly in CoCo Desktop
- Identify and mask PII using Snowflake governance features

Prerequisite: CoCo Desktop must be installed before the session. Each participant will work in an isolated sandbox database created from `CURRENT_USER()` with a dedicated HOL role.

Sandbox Environment

Isolated per user: Each participant gets a dedicated HOL database created automatically based on `CURRENT_USER()`.

Scoped permissions: Your HOL role can only read and write to your own HOL schema — no risk of impacting shared environments.

Teardown: Databases are temporary and can be dropped after the session.

Pre-requisite: Ensure CoCo Desktop is installed and can connect to your Snowflake account before Oct 14.

Session Agenda

Time	Min	Activity	Description
0:00	30	CoCo Desktop Overview	Navigation, best practices, and key capabilities walkthrough
0:30	30	Lab 01: Explore Data (Guided)	Instructor-led walkthrough of data exploration in CoCo Desktop
1:00	10	*— Break —*	
1:10	30	Self-Paced Lab — Round 1	Choose any lab from Labs 02–12
1:40	10	*— Break —*	
1:50	30	Self-Paced Lab — Round 2	Choose a second lab from Labs 02–12
2:20	30	Self-Paced Lab — Round 3 (Optional)	Choose a third lab or revisit a previous one
2:50	10	Wrap-Up & Q&A	Review key takeaways, share experiences, and open Q&A

How it works: After the guided overview and Lab 01, you choose your own adventure. Pick 2–3 labs that match your role and interests. Each self-paced lab is designed to take ~30 minutes.

Guided Session — 60 Minutes

COCO DESKTOP OVERVIEW & BEST PRACTICES

30 MIN | INSTRUCTOR-LED

Get oriented in CoCo Desktop and learn the patterns that make AI-assisted development effective.

- Launch CoCo Desktop and connect to your Snowflake sandbox environment
- Tour the interface — file explorer, chat panel, editor, and terminal
- Best practices for prompting: be specific, provide context, iterate
- Understanding how CoCo Desktop reads your project context (files, schemas, metadata)
- Managing chat sessions — when to start fresh vs. continue
- Quick demo of key workflows: code generation, explain, refactor, and debug

LAB 01 — EXPLORE DATA

30 MIN | GUIDED — EVERYONE TOGETHER

Walk through a hands-on data exploration exercise as a group, led by the instructor.

- Connect CoCo Desktop to your personal sandbox HOL database
- Ask CoCo to describe tables, show sample rows, and explain column semantics
- Generate exploratory SQL queries using natural language
- Profile data distributions, identify nulls, and spot anomalies
- Visualize results and iterate on your questions
- **Outcome:** Comfortable navigating data in CoCo Desktop independently

Self-Paced Lab Catalog — Choose 2–3 Labs (~30 Min Each)

After the guided session, pick labs that match your role and interests. Each lab is self-contained and takes approximately 30 minutes.

Developer | Data Engineer | Data Scientist | All Roles | Governance

Lab 02 — Build a Pipeline

Data Engineer

Design and build a data ingestion and transformation pipeline using CoCo Desktop. Create staging tables, write ELT logic, and schedule tasks.

Lab 03 — Data Quality Checks

Data Engineer

Build automated data-quality checks: null detection, uniqueness constraints, range validation, and referential integrity tests.

Lab 04 — Generate Diagrams

All Roles

Use CoCo to generate ERD, data-flow, and architecture diagrams from your schema metadata and pipeline definitions.

Lab 05 — Build a Semantic View

Data Engineer

Create a Semantic View that maps business terminology to SQL, enabling natural-language querying for non-technical users.

Lab 06 — Build a Cortex Agent

Developer

Build a Cortex Agent that answers domain-specific questions using your H&W data, including tool use and retrieval-augmented generation.

Lab 07 — Notebook Forecasting

Data Scientist

Create a Snowflake Notebook for time-series forecasting. Load data, build a model, generate predictions, and visualize results.

Lab 08 — Author a CoCo Skill

All Roles

Build a reusable CoCo Skill that encapsulates a common workflow — from prompt design to testing and deployment.

Lab 09 — Fix a Slow Query

Data Engineer

Diagnose a slow-running query using CoCo. Analyze the query profile, identify bottlenecks, and apply optimization techniques.

Lab 10 — Ship a Streamlit App

Developer

Build and deploy a Streamlit in Snowflake app — interactive UI, data connections, and publishing — all within CoCo Desktop.

Lab 11 — Find and Mask PII

Governance

Use CoCo to scan tables for personally identifiable information, then apply Snowflake masking policies to protect sensitive data.

Lab 12 — Cost and Usage Capstone

All Roles

Capstone exercise: analyze Snowflake cost and usage data, build dashboards, identify optimization opportunities, and present findings.

Lab Details — Data Engineering & Governance

LAB 02 — BUILD A PIPELINE

~30 MIN SELF-PACED

Design an end-to-end data pipeline using AI-assisted SQL generation in CoCo Desktop.

- Create staging and target tables with CoCo-generated DDL
- Write transformation logic using CTEs, window functions, and aggregations
- Build Snowflake Tasks for scheduled pipeline execution
- Add error handling and logging patterns
- **Outcome:** A working ELT pipeline ready for scheduling

LAB 03 — DATA QUALITY CHECKS

~30 MIN SELF-PACED

Build a suite of automated data-quality validations powered by CoCo.

- Generate null-check, uniqueness, and range-validation queries from natural language
- Build referential integrity tests across related tables
- Create a reusable DQ framework with pass/fail reporting
- Schedule DQ checks as downstream tasks in your pipeline
- **Outcome:** An automated data-quality test suite

LAB 05 — BUILD A SEMANTIC VIEW

~30 MIN SELF-PACED

Create a business-friendly Semantic View that enables natural-language queries over your data.

- Select base tables and define dimensions, measures, and time grains
- Add business descriptions and synonyms for column names
- Define relationships between tables (joins and keys)
- Test the Semantic View by asking CoCo business questions against it
- **Outcome:** A published Semantic View usable by analysts via CoCo

LAB 09 — FIX A SLOW QUERY

~30 MIN SELF-PACED

Diagnose and optimize a slow query using CoCo's explain and optimization capabilities.

- Run a poorly-performing query and capture its query profile
- Ask CoCo to explain the execution plan and identify bottlenecks
- Apply optimizations: clustering keys, filter pushdown, join reordering
- Compare before/after performance metrics
- **Outcome:** A measurably faster query with documented optimizations

LAB 11 — FIND AND MASK PII

~30 MIN SELF-PACED

Identify and protect personally identifiable information using CoCo and Snowflake governance.

- Ask CoCo to scan table schemas and sample data for potential PII fields
- Review and classify detected PII (names, emails, SSNs, phone numbers, etc.)
- Generate and apply Snowflake dynamic data masking policies
- Validate that masking is enforced for non-privileged roles
- **Outcome:** PII fields identified and masked with governance policies

Lab Details — Developer, Data Science & General

LAB 04 — GENERATE DIAGRAMS

~30 MIN SELF-PACED

Automatically generate visual diagrams from your Snowflake metadata using CoCo.

- Generate entity-relationship diagrams (ERDs) from table schemas
- Create data-flow diagrams for your pipeline architecture
- Produce Mermaid or PlantUML diagram code via natural-language prompts
- Iterate on layouts, labels, and groupings through conversation
- **Outcome:** Exportable architecture and data-model diagrams

LAB 06 — BUILD A CORTEX AGENT

~30 MIN SELF-PACED

Build an AI agent that can answer domain-specific H&W questions using your Snowflake data.

- Define the agent's purpose, tools, and data access scope
- Configure retrieval-augmented generation (RAG) with Cortex Search
- Add SQL tool access so the agent can query tables on demand
- Test the agent with sample H&W questions and refine its behavior
- **Outcome:** A working Cortex Agent deployed on Snowflake

LAB 07 — NOTEBOOK FORECASTING

~30 MIN SELF-PACED

Build a time-series forecasting workflow inside a Snowflake Notebook.

- Create a Notebook and load historical data via SQL cells
- Use Cortex ML forecasting functions or Python-based models
- Generate predictions and visualize forecast results
- Document your methodology with Markdown cells
- **Outcome:** A shareable Notebook with a working forecast model

LAB 08 — AUTHOR A COCO SKILL

~30 MIN SELF-PACED

Create a reusable CoCo Skill that automates a repetitive technical workflow.

- Identify a repeatable pattern in your daily work (report, check, transformation)
- Write and test the Skill's prompt template with parameterized inputs
- Validate the Skill produces consistent, correct outputs
- Share the Skill with teammates for team-wide productivity
- **Outcome:** A published, reusable CoCo Skill

LAB 10 — SHIP A STREAMLIT APP

~30 MIN SELF-PACED

Build and deploy an interactive Streamlit in Snowflake application using CoCo Desktop.

- Scaffold a Streamlit app with CoCo-generated boilerplate code
- Add interactive widgets: filters, dropdowns, date pickers
- Connect to Snowflake data and render charts and tables
- Deploy the app to Snowflake for team access
- **Outcome:** A live Streamlit app running in Snowflake

LAB 12 — COST AND USAGE CAPSTONE

~30 MIN SELF-PACED

Capstone exercise that ties together multiple skills to analyze Snowflake cost and usage.

- Query Snowflake's ACCOUNT_USAGE views for credit consumption and warehouse metrics
- Build cost-allocation breakdowns by warehouse, user, and query type
- Identify optimization opportunities (idle warehouses, expensive queries)
- Create a summary dashboard or report of findings
- **Outcome:** A cost-analysis deliverable with actionable recommendations

Tips for Success

Tip 1 — Start with Lab 01: The guided Explore Data lab builds the foundation for every other lab. Don't skip it even if you're experienced — it covers CoCo Desktop–specific patterns.

Tip 2 — Pick Labs by Role: Data Engineers should prioritize Labs 02, 03, 05, and 09. Developers should try Labs 06, 08, and 10. Data Scientists should focus on Labs 07 and 12. Labs 04, 08, and 12 are great for everyone.

Tip 3 — Give CoCo Context: When prompting, mention your table names, schema, and what you're trying to achieve. The more specific you are, the better the generated code.

Tip 4 — Always Review Generated Code: CoCo is powerful but not infallible. Read the generated SQL, Python, or Streamlit code before running it — treat it as a smart first draft.

Tip 5 — Use Your Sandbox Freely: Your HOL database is isolated and disposable. Experiment without fear — create tables, run transforms, and break things. That's what it's for.

Next Steps After Dev Day

Revisit Labs: Your sandbox environment remains available after the session. Go back and complete labs you didn't get to.

Build a Skill: Take a repetitive pattern from your real work and author a CoCo Skill within 14 days.

Semantic Views: Work with your team to build Semantic Views for your top H&W datasets within 30 days.

Ship Something: Deploy at least one Streamlit app, Cortex Agent, or automated pipeline to production within 30 days.

Feedback: Share your experience and suggestions with the training team to improve future Dev Day sessions.